Literature Applied

Which paper changed which line

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1 Part 7 — What we took from the literature, and where it shows up

The honest question is not ”did you cite papers”, it is ”would this project be different if you had not read them”. This file answers that, one paper at a time, with the line of code or the experiment that changed. Where a paper made a claim we could test, we tested it, and we say what happened — including the two places where we got it wrong first.

1.1 7.1 The short version

Paper	What we took from it	Where it lives	Did we test its claim?
Kennedy & Eberhart 1995	the velocity update itself	`pso_wifi_placement.py` §3	— (it is the algorithm)
Shi & Eberhart 1998	inertia weight `w`: 0.9 → 0.4	`optimize()`	yes, via the diversity curve
Kennedy & Mendes 2002	**swarm topology is a variable, not a given**	`topology` / `_social_attractor()`	**yes — and it corrected us**
Wolpert & Macready 1997 (NFL)	you must *demonstrate* fit to the landscape, not assert it	the entire ablation	it is why the ablation exists
Rappaport 2002	log-distance path loss + wall attenuation	`WifiFloorProblem.fitness()`	— (it is the objective)
Bellman 1957 / Howard 1960	value iteration; the contraction bound	`value_iteration()`	yes — measured rate = 0.9900 = γ
Watkins & Dayan 1992	the TD update; the `max` that makes it off-policy	`q_learning()`	— (it is the algorithm)
Jaakkola et al. 1994 / Tsitsiklis 1994	”every (s,a) infinitely often”	exploring starts	yes — off = 76.7% vs on = 25.1%
Even-Dar & Mansour 2003	α = 1/(1+n)^0.7, and *why constant α cannot work*	`alpha_omega`	**yes — constant α stalls at 50.4%**

Paper	What we took from it	Where it lives	Did we test its claim?
Agarwal et al. 2020 + Li et al. 2024	**the sample-complexity gap that explains our headline result**	`certainty_equivalence()`	yes — predicted gap, observed gap
Sutton 1990 (Dyna)	the model-free/model-based boundary is a spectrum	framing of Part B	noted as future work

Two of these did not merely inform the project. They **changed a result we had already written down.** Those two are worth the rest of this document.

1.2 7.2 Kennedy & Mendes (2002) — the paper that caught us over-claiming

What we had. Our first ablation was binary: communication on (every particle pulled towards the single global best) or off (`c2 = 0`). On, 89.91. Off, 87.30 — indistinguishable from random search. Communication matters. Done.

What the paper says. Kennedy & Mendes systematically compare 70 different swarm communication topologies, and their finding is narrower and more useful than the folklore:

> *"greater connectivity speeds up convergence, though it does not tend to improve the population's ability to discover global optima"*

Concretely, they measure the fully-connected `gbest` swarm hitting their criterion in a median of **404.8 iterations but only 85% of the time**, against a ring `lbest` swarm needing **755.5 iterations but succeeding 94% of the time**.

They also warn about the other end, which is exactly our `c2 = 0` row:

> *"inhibiting communication too much results in inefficient allocation of trials, as individual particles wander cluelessly through the search space"*

What we did with it. We implemented a ring topology (`ring_k` neighbours each side, information walks around the circle one hop per iteration instead of teleporting) and re-ran the ablation at the same 3,030-evaluation budget.

What we got, and how we got it wrong first. Our first pass reported that the ring was simply *better* — mean 89.99 against gbest's 89.91. A unit test asserting that failed on a different set of seeds, which was the correct outcome. We then tested it properly, on 30 paired seeds:

	mean	sd	stagnates at
fully connected (gbest)	89.831	0.317	iteration **83** / 100
ring, k = 1	89.961	**0.129**	iteration **100** / 100

- **Wilcoxon on the mean: p = 0.92.** Not significant. The ring does **not** find a better optimum.
- **F-test on the variance: 6× smaller, $p = 6 \times 10^{-6}$.** Highly significant.
- The fully-connected swarm **gives up at iteration 83**. The ring is still improving when the budget runs out.

That is the paper's claim, reproduced: more connectivity converges *faster*, not *better*. The gain from the ring is **reliability**, not peak performance. Our first write-up said "better" and the statistics did not support it.

The bit we would have got wrong if we had only skimmed it. The paper does *not* recommend the ring. It ranks ring-with-self **64th of the 70** topologies tested ("slow and inaccurate") and recommends the **von Neumann** lattice instead. So we must not say "the literature says use a ring". We say the narrower true thing: the literature predicts a speed-versus-reliability trade between dense and sparse topologies, and on our landscape that trade appears exactly as described. Trying von Neumann is the obvious next experiment and we have not run it.

1.3 7.3 Even-Dar & Mansour (2003) — a prediction we could falsify

The Robbins–Monro conditions require $\Sigma\alpha = \infty$ and $\Sigma\alpha^2 < \infty$. A **constant** α satisfies the first and fails the second, so it provably converges only to a *neighbourhood* of Q^*, never to Q^* itself. A polynomial rate $\alpha_n = 1/(1+n)^\omega$ with $\omega \in (0.5, 1]$ satisfies both.

This is a falsifiable prediction about *our* problem, so we put both in the sweep rather than just citing the theorem:

α schedule	regret
polynomial, $\omega = 0.7$	**25.1%**
polynomial, $\omega = 0.5$	28.2%
constant $\alpha = 0.10$	50.4%
constant $\alpha = 0.50$	57.9%

The constant-α runs stall exactly as the theory says they must. We did not have to take the theorem on faith, and neither does the reader.

1.4 7.4 Agarwal et al. (2020) and Li et al. (2024) — why our headline result happens

This is the one that turned an empirical curiosity into an explanation.

Our result. Take the 720,000 transitions Q-learning consumed, estimate the maximum-likelihood model $\hat{P}$ from them, and plan on it. That *certainty-equivalence* planner reaches **1.43%** regret against the Q-learner's **15.13%** — a factor of ten, on identical data. We could describe this but not explain it.

The near-miss. Our first instinct was to cite Azar, Munos & Kappen (2013), which proves a minimax bound on model-based value iteration. When we checked what it actually proves, it does **not** support our claim: its lower bound is information-theoretic and binds *every* algorithm, model-free included. Citing it for a model-based-beats-model-free claim would have been wrong, and an examiner who knows the paper would have said so.

The papers that do support it. The separation is real, and it is sharper than we expected:

- **Agarwal, Kakade & Yang (2020)** prove the *plug-in* (certainty-equivalence) estimator — build $\hat{P}$, plan optimally in the empirical MDP — is **minimax optimal**, achieving $\Theta(|S||A| / ((1-\gamma)^3\varepsilon^2))$.
- **Li, Cai, Chen, Wei & Chi (2024)** settle vanilla synchronous Q-learning at $\Theta(|S||A| / ((1-\gamma)^4\varepsilon^2))$ and state plainly that this "unveils the strict sub-optimality of Q-learning when $|A| \geq 2$".

So certainty-equivalence sits at $1/(1-\gamma)^3$ and vanilla tabular Q-learning is stuck at $1/(1-\gamma)^4$ — **a full factor of $1/(1-\gamma)$ worse in the effective horizon.**

Now put our numbers in. We chose $\gamma = 0.99$, so $1/(1-\gamma) =$ **100**. The theory predicts our Q-learner should be roughly two orders of magnitude behind in sample efficiency, purely because of the horizon. That is not a curiosity about our hall. It is a known, proved separation, and our 1.43%-versus-15.13% is what it looks like from the inside.

The caveat we must not drop. The separation is against *vanilla* Q-learning, not model-free RL in general. Variance-reduced Q-learning variants do reach the optimal rate. So the correct sentence is "tabular Q-learning is provably sample-inefficient here", **not** "model-free learning is worse". We say the first.

The mechanism, in one line. A Q-learning update spends a transition on one cell of the table and discards it. A learned model *stores* it, and every subsequent Bellman backup re-reads it — so one sample propagates across the whole state space instead of into the single cell where it landed.

1.5 7.5 Wolpert & Macready (1997) — the paper that shaped the method, not the code

No Free Lunch says that averaged over all objective functions, every optimiser is identical. The practical consequence is that "PSO is good" is not a claim you can make; only "PSO is good *on this landscape*, and here is the evidence" is.

That single idea is why Part A is built the way it is. It is why we do not report "PSO found a good layout" and stop; it is why there is a matched-budget random search, a matched-budget grid search, a swarm-size sweep, a communication ablation and a topology comparison. Every one of those exists to answer "compared to what?".

It is also why the Part B result survives. We went looking for evidence that model-free learning wins, ran the experiment that could embarrass us, and it did. Reporting that is the same discipline.

1.6 7.6 What we read and did not use

Being straight about this is part of the point.

- **The twelve applications in PART2** are a genuine literature survey with verified citations, but not one of them changed a design decision here. They are context, not input. If we were doing it again we would mine them for *encoding* tricks (several of those papers are interesting precisely because of how they encode a constraint into the representation rather than penalising it) and see whether our AP-position encoding could absorb the separation constraint structurally instead of as a penalty term.
- **The pump-scheduling RL papers** (Hajgató et al. 2020; Hu et al. 2023; Pei et al. 2025) all use deep RL on large water-distribution networks. We deliberately went the other way — small enough that value iteration gives exact ground truth, so every other method can be scored against the true optimum rather than against another approximation. That is a design decision *informed* by them, but it is a decision to do the opposite.
- **Clerc & Kennedy (2002)** — the constriction factor $\chi = 0.7298$ is an alternative to the inertia weight, and Eberhart & Shi (2000) found it performs better on their benchmarks (while still keeping velocity clamping, contrary to a common misreading). We use the inertia weight because it is what the course teaches and what the slide shows. Comparing the two on our landscape is a clean experiment we did not run.
- **Sutton (1990), Dyna** — Dyna interleaves real experience with simulated experience from a learned model, and sits exactly on the boundary our Part B is about. Our certainty-equivalence arm is the batch, one-shot version of the same idea. Dyna-Q is the incremental version and would be the natural third arm.

1.7 7.7 If he asks "which paper actually changed your mind?"

Two, and say both.

Kennedy & Mendes (2002) turned our communication ablation from a yes/no question into a "how much, and in what structure" question — and then corrected us when we over-read our own numbers. The mean difference between the ring and the fully connected swarm is not significant. The variance difference is, by six times. We had written "better" and had to change it to "steadier".

Li et al. (2024) with **Agarwal et al. (2020)** gave us the reason our headline result happens at all. We had the measurement — a learned model beats Q-learning by ten times on identical samples — and no explanation. The explanation is a proved sample-complexity separation of $1/(1-\gamma)$, which at our $\gamma = 0.99$ is a factor of a hundred. We also learned, checking it, that the paper we *nearly* cited for this does not support the claim, which is its own lesson about reading past the abstract.